

95-868 Exercises 6

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Here are some libraries that you may need.

```
library(ggplot2)
library(plyr)
library(reshape2)
```

```
## Warning: package 'reshape2' was built under R version 4.5.3
```

```
library(knitr)
library(binom) # for binom.confint()
```

```
## Warning: package 'binom' was built under R version 4.5.3
```

We will use the data files `county_data.csv` and `county_ggplot.rda`, which were part of the lecture materials.

```
county.data = read.csv(file = 'county_data.csv', header=TRUE)
load('county_ggplot.rda')
```

Problem 1:

Using `binom.confint()`, compute 95% confidence intervals for the per capita rate of infant deaths in each county. Set the argument `method = "exact"` in your function call.

```
# Compute 95% confidence intervals using the exact method
county.conf = binom.confint(x = county.data$infant.deaths,
                           n = county.data$population,
                           methods = "exact")

# Combine the confidence intervals back with the county data
county.data = cbind(county.data, county.conf)
```

Problem 2:

What are the counties with the top ten lower bounds for the infant death per capita rate? Make a table using the `kable()` command. Be sure to choose the number of digits shown in the table appropriately.

```
# Arrange by lower bound in descending order
county.top.lower = arrange(county.data, desc(lower))

# Display top 10 using kable
kable(subset(county.top.lower[1:10, ],
             select = c("name", "population", "infant.deaths", "lower", "upper")),
      digits = 5)
```

name	population	infant.deaths	lower	upper
Tunica, MS	10491	9	0.00039	0.00163
Crockett, TX	3782	4	0.00029	0.00271
Allendale, SC	10418	7	0.00027	0.00138

name	population	infant.deaths	lower	upper
Quitman, GA	2649	3	0.00023	0.00331
Norfolk, VA	236051	68	0.00022	0.00037
Wilkinson, GA	10031	6	0.00022	0.00130
Orangeburg, SC	90615	27	0.00020	0.00043
Muscogee, GA	186613	48	0.00019	0.00034
Bibb, GA	154891	40	0.00018	0.00035
Hinds, MS	248959	60	0.00018	0.00031

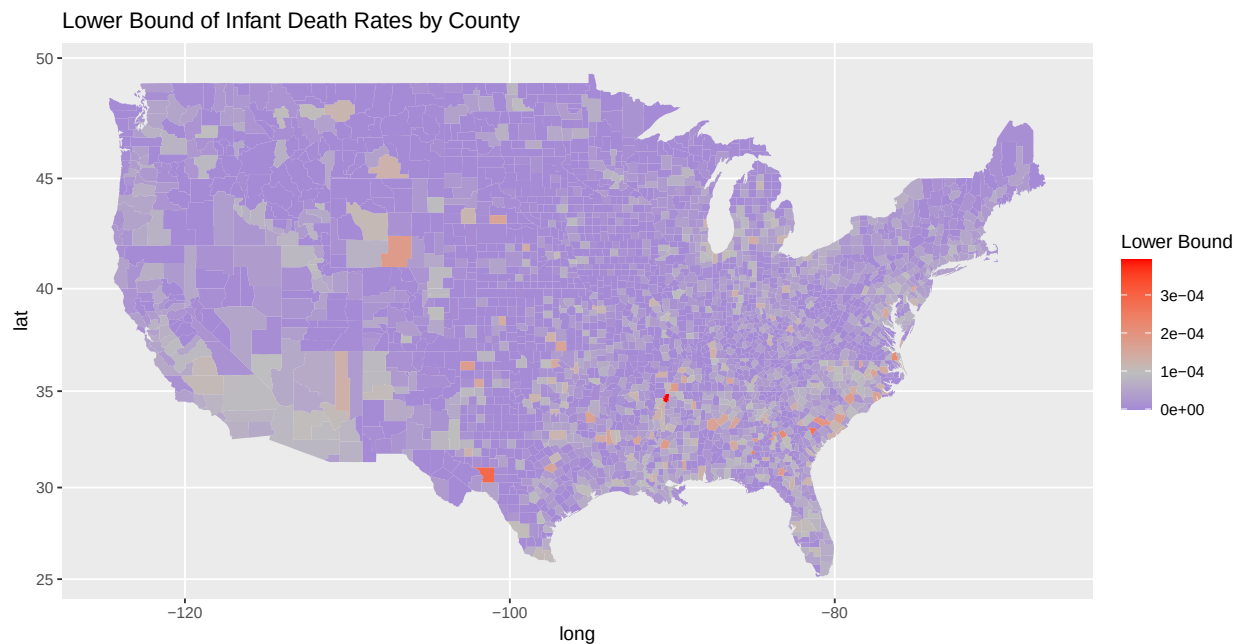
Problem 3:

Make a choropleth plot showing the lower bound on the per capita infant deaths. Choose the color scale so that the midpoint is at the US per capita rate for infant deaths.

```
# Calculate nationwide per capita rate for the midpoint
us.infant.rate = sum(county.data$infant.deaths) / sum(county.data$population)

# Merge with map data
county.mapdata = merge(county.gg, county.data, by.x = 'fips', by.y = 'STCOU')
county.mapdata = arrange(county.mapdata, order)

# Create the plot
ggplot(data = county.mapdata, aes(x = long, y = lat, group = group, fill = lower)) +
  geom_polygon() +
  scale_fill_gradient2(low = 'blue', mid = 'grey', high = 'red', midpoint = us.infant.rate) +
  coord_map() +
  labs(title = 'Lower Bound of Infant Death Rates by County', fill = 'Lower Bound')
```



Problem 4:

Use 95% confidence intervals to identify the counties with extremely LOW per capita rates of infant deaths. (We've only looked at counties with high rates so far. You may need to think about how to do this)

Make a table showing your top ten selections.

```
options(scipen = 999)
# Arrange by upper bound in ascending order (lowest "maximum" risk)
county.low.rates = arrange(county.data, upper)

# Display top 10 counties with the lowest upper bounds
kable(subset(county.low.rates[1:10, ],
  select = c("name", "population", "infant.deaths", "lower", "upper")),
  digits = 5)
```

name	population	infant.deaths	lower	upper
Sarasota, FL	368841	8	0.00001	0.00004
Somerset, NJ	320745	8	0.00001	0.00005
Bergen, NJ	885664	32	0.00002	0.00005
San Luis Obispo, CA	261706	7	0.00001	0.00006
Monmouth, NJ	641036	24	0.00002	0.00006
Middlesex, MA	1475519	65	0.00003	0.00006
Douglas, CO	271955	8	0.00001	0.00006
Saratoga, NY	216259	6	0.00001	0.00006
New York, NY	1625251	80	0.00004	0.00006
Norfolk, MA	656122	28	0.00003	0.00006